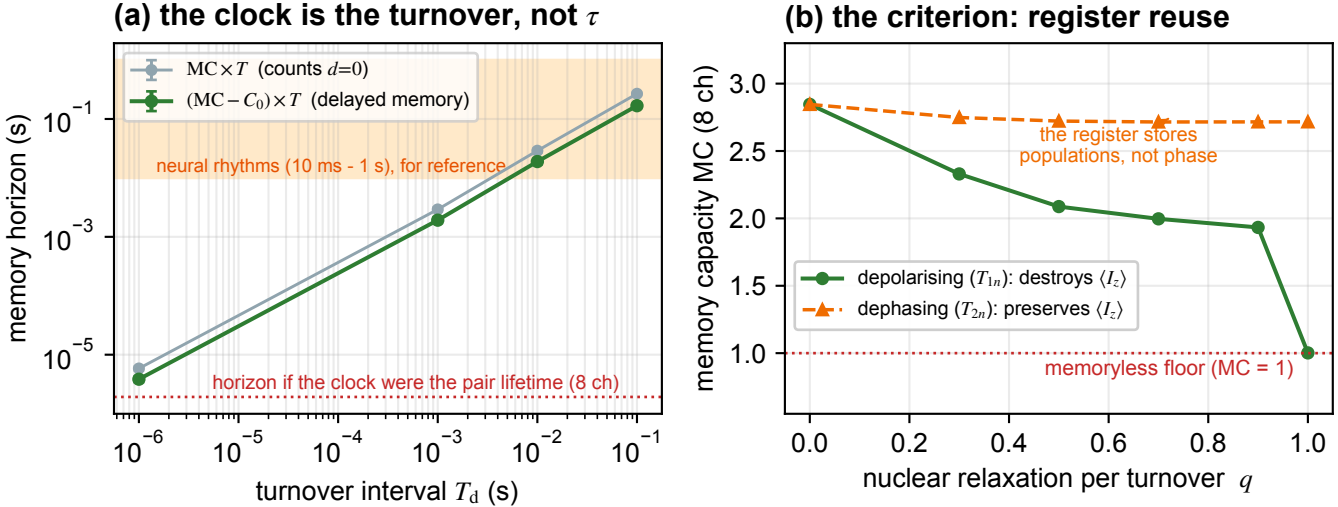
(a) the clock is the turnover, not τ
MC × T (counts d=0)
(MC − C_0) × T (delayed memory)
neural rhythms (10 ms - 1 s), for reference
horizon if the clock were the pair lifetime (8 ch)
memory horizon (s)
10^{-1}
10^{-3}
10^{-5}
10^{-6}
10^{-5}
10^{-4}
10^{-3}
10^{-2}
10^{-1}
turnover interval T_d (s)
(b) the criterion: register reuse
memory capacity MC (8 ch)
3.0
2.5
2.0
1.5
1.0
the register stores
populations, not phase
depolarising (T_{1n}): destroys $\langle I_z \rangle$
dephasing (T_{2n}): preserves $\langle I_z \rangle$
memoryless floor (MC = 1)
0.0
0.2
0.4
0.6
0.8
1.0
nuclear relaxation per turnover q